Domain Augmentation for Better Generalization (CIFAR-10 $\rightarrow$ CIFAR-10-C)

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Final Project Presentation

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Goal : Improve CNN robustness to real-world corruptions using a progressive augmentation curriculum.

Problem Statement & Description

Problem: CNNs trained on clean images generalize poorly under **domain shift** (noise, blur, weather, compression).

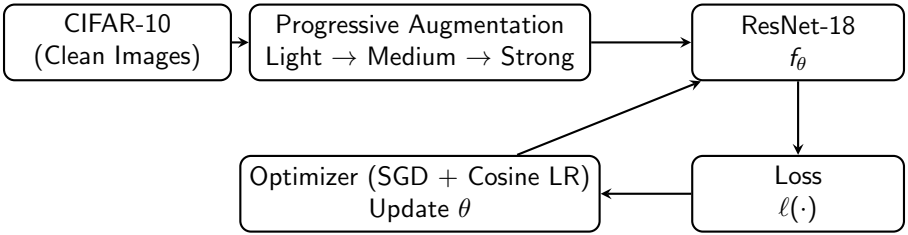
Setting:

- **Train:** CIFAR-10 (50k clean images)
- **Test:** CIFAR-10-C (15 corruptions $\times$ 5 severities)

Why it matters: Real deployments face corruptions not seen during training (camera noise, motion blur, low light, JPEG artifacts).

Baseline model: ResNet-18 (efficient, standard robustness benchmark).

Proposed Solution Framework



Idea: The model is trained using a curriculum-style augmentation strategy, starting with light distortions and gradually increasing difficulty to improve robustness.

Mathematical Formulation & Training Schedule

Training objective (curriculum augmentation):

$$\min_{\theta} \frac{1}{E} \sum_{t=1}^{E} \mathbb{E}_{(x,y)\sim\mathcal{D}} \Big[\ell\big(f_\theta(T_{\alpha(t)}(x)),\, y\big) \Big]$$

- $T_{\alpha(t)}$: augmentation operator with strength $\alpha(t)$ that increases with epoch t
- f_θ: ResNet-18, ℓ: cross-entropy

Stepwise schedule (main):

- Epochs 1–5: **Light** (small jitter, slight blur/noise)
- Epochs 6–10: **Medium** (moderate jitter, moderate blur/noise)
- Epochs 11–20: **Strong** (strong blur/noise, strong contrast changes)

Baselines: No-Aug.

Metrics

Final goal: Achieve improved robustness on CIFAR-10-C compared to baseline models, while maintaining competitive clean CIFAR-10 accuracy.

Evaluation metrics:

- **Mean Corruption Accuracy (mCA):**

$$\text{mCA} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{5} \sum_{s=1}^{5} \text{Acc}(c, s)$$

- **Mean Corruption Error (mCE):**

$$\text{mCE} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{\text{Err}_{\text{model}}(c)}{\text{Err}_{\text{baseline}}(c)}$$

Training & Test Results (Clean CIFAR-10)

Setup:

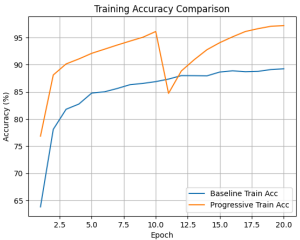
- Model: ResNet-18
- Dataset: CIFAR-10
- Comparison: Baseline vs PDA

Key Observations:

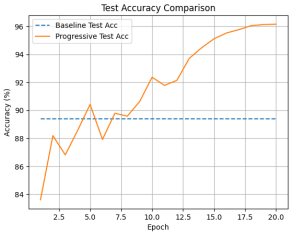
- Baseline converges faster initially
- PDA shows slower early learning
- PDA generalizes better on test data

Accuracy Curves on Clean

Training Accuracy



Test Accuracy



Clean & Corrupted Test Results (CIFAR-10)

Evaluation Setup:

- Dataset: CIFAR-10 (test)
- Corruptions: CIFAR-10-C (15 types, severities 1–5)

Key Takeaways:

- PDA improves clean accuracy
- PDA is significantly more robust to corruptions
- Lower mCE indicates stronger robustness

Performance Comparison

Method	Clean Acc (%)	MCA (%)	mCE (%)
Baseline	88.73	66.88	33.12
PDA	**95.79**	**77.17**	**22.83**

Higher MCA and lower mCE indicate improved corruption robustness.

Comparison with Prior Work (CIFAR-10-C, VGG16)

Paper Results (Error %, VGG16)

Method	Noise	Blur	Weather	Digital
Natural	46.22	30.34	17.99	23.64
PDA-3-1.0	**18.88**	**18.05**	**17.90**	**20.18**

Our Results (Error %, from Accuracy)

Method	Noise	Blur	Weather	Digital
Natural	70.79	29.11	15.54	24.57
PDA	**62.74**	**26.11**	**13.46**	**24.04**

Error computed as 100 − Accuracy. Lower is better.

Observation:

- PDA consistently reduces corruption error compared to natural training
- Largest improvement observed for noise corruptions
- Trends align with prior work despite different absolute values

Challenges & Limitations

Experimental Limitations:

- Evaluation performed on CIFAR-10-C only (limited image resolution).
- Absolute corruption error values differ from paper due to:
  - different training schedules,
  - implementation-level differences.
- No hyperparameter sweep for PDA strength (fixed configuration).

Methodological Constraints:

- Not even a glance of noise introduced during training , and model failed .
- Training time span, many epochs need too much time and hence limitided training epochs.

Conclusion

Summary of Findings:

- Progressive Data Augmentation (PDA) improves robustness to common corruptions.
- PDA achieves higher clean accuracy compared to natural training.
- Corruption error reductions are consistent with trends reported in prior work.

Key Takeaway:

- Simple augmentation strategies can significantly improve model robustness without sacrificing clean performance.

Future Work:

- Extend evaluation to ImageNet-C and higher-resolution datasets.
- Combine PDA with adversarial training for joint robustness.